Chapter 11

Immunology

Learning Objectives

After studying this chapter the students will be able to,

- *Understand the Antigen Antibody reactions*
- *Know the principle behind Western Blot techniques*
- *Learn about Hypersensitivity*
- *Gain knowledge about Transplantation*
- *Know Immunization/Vaccination*
- *Appreciate the Updated National Immunization Schedule chart.*

Chapter Outline

11.1 Antigen Antibody Reactions

11.2 Western Blot Techniques

11.3 Hypersensitivity

11.4 Transplantation

11.5 Immunization/Vaccination

11.6 Updated National Immunization Schedule Chart

The immune system refers to a collection of cells and proteins that function to protect the skin, respiratory passages, intestinal tract and other areas from foreign antigens, such as microbes (organisms such as bacteria, fungi, and parasites), viruses, cancer cells, and toxins. The immune system works by recognising the difference between one's own body cells and alien cells, allowing it to destroy anything that could be potentially harmful. Immune deficiency diseases decrease the body's ability to fight invaders, causing vulnerability to infections. In the previous year, we have elaborately discussed with the main components and function of the immune system. This chapter deals with the role of Immune system in both health and disease.

11.1 Antigen Antibody Reactions

The interaction between antigen and antibody is called antigen-antibody reactions. It is abbreviated as Ag-Ab reaction. This reaction is the basis of humoral immunity. The antigen and the antibody react to form immune complex.

Ag + Ab ---------- Ag – Ab complex

The reaction between antigen and antibody is highly specific. It is compared to the lock and key system. The part of the antigen that combines with the antibody is called epitope or antigenic determinant. The part of antibody which combines with the antigen is called paratope or antigen

TN_GOVT_XII_Micro_Biology_CH11.indd 230 04-03-2019 12:31:57